Chenghao Yang

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PROFILE

GIS professional (MSc, UBC) with 6+ years of experience delivering geospatial solutions for environmental, Indigenous land stewardship, and infrastructure projects. Specializes in enterprise GIS administration, spatial analysis, and end-to-end digital delivery — from field data collection through to web maps, dashboards, and client reporting. Contributes to proposal development, technical mentorship, and organizational capacity-building alongside hands-on digital delivery. BC Class 5 driver's license.

TECHNICAL SKILLS

GIS Software: ArcGIS Pro, ArcGIS Online, Portal for ArcGIS, ArcGIS Enterprise (license management, user configuration, hosted feature layer administration), QGIS, ENVI, SAGA

Programming & Automation: Python, R, Arcade, FME, ModelBuilder, Power Automate

Data & Governance: Geodatabase schema design (Domains/Subtypes), CAD-to-GIS integration, SQL Server (cross-organization data acquisition), metadata standards, version control

Field & Web: Survey123, FieldMaps, ArcGIS Dashboards, ArcGIS Experience Builder, Leaflet

Spatial Analysis: LiDAR point cloud processing, DSM/DTM/CHM derivation, terrain and hydrological modelling, volumetric estimation, raster/vector analysis

WORK EXPERIENCE

GIS Specialist · Tapestry Research Group · Vancouver, BC *2023 – Present*

Infrastructure & Spatial Analysis

- Led 3D Airport OLS modeling across five threshold scenarios per regulatory standards; processed LiDAR to derive CHM raster, identified hazard trees, and estimated harvest volumes to support infrastructure compliance.
- Built a Python pipeline to automate hydrological analysis across 27 LiDAR-derived DSM sites at variable thresholds as part of a bioregional atlas mapping project; maintained metadata records and performed QA across territory, landscape, and site scales.
- Developed end-to-end LiDAR workflow for quarry stockpile volumetrics: point cloud cleaning, ground classification, and volume calculation; authored repeatable process documentation and client reports.

Digital Delivery & Systems

- Designed and administered organization-wide geodatabase schemas and CAD-to-GIS transformation standards; managed ArcGIS Online/Enterprise environments including user licensing, hosted feature layers, credit monitoring, and dataset archiving.
- Designed Survey123 forms and backend data schemas; managed full field monitoring data lifecycle across multiple projects including hosted feature layer archiving, credit control, and extraction into geodatabases.
- Delivered asset management system for a First Nations community under a federally-funded program: built Survey123 data collection, ArcGIS Dashboards with Arcade widgets and client website integration, and Power Automate workflows for incident reporting.
- Built web maps and applications using ArcGIS Experience Builder to support digital project deliverables.
- Queried SQL Server databases from partner organizations to acquire and integrate external geospatial datasets.

Project Delivery & Team

- Designed and produced multi-theme map atlases and netdown assessment workflows using advanced cartographic principles; delivered final printed booklets to support Indigenous consultation and executive decision-making.
- Contributed to client proposals and service agreements through task-level effort estimation and scope development.
- Designed organization-wide map review template for QA/QC; onboarded three junior GIS staff through hands-on training and internal documentation.

Business Analyst · China Telecom · Vancouver, BC *2019 – 2022*

Supported client operations, stakeholder coordination, and project reporting during undergraduate studies. Developed ArcGIS StoryMap to visualize partner locations across the region for management handover.

EDUCATION

MSc — Geomatics for Environmental Management *2023*
University of British Columbia

BSc — Environmental Science (Environmetrics) · **GIS Certificate** *2022*
Simon Fraser University

PROFESSIONAL DEVELOPMENT

G.I.T. Applicant (EGBC) · Safe Software FME Academy (In Progress) · Invited Attendee — International Cartographic Conference 2025